# Adaptation Experiment 1: Vanilla Backbone Reference

## 1 Theoretical overview and goal

A forecasting adapter is useful only if it improves a well-defined, untouched backbone. For a look-back $x \in \mathbb{R}^L$, a pretrained forecaster produces $\hat{y}_0 = f(x) \in \mathbb{R}^H$. The univariate control evaluates $f$ directly, without retrieved neighbors, gates, or trainable adaptation. It therefore separates backbone error from error introduced by the retrieval and adaptation pipeline.

The main metrics are

$$\mathrm{MSE} = H^{-1}\|\hat{y}_0 - y\|_2^2, \qquad \mathrm{nMSE} = H^{-1}\left\|\frac{\hat{y}_0 - y}{\sigma(x)+\varepsilon}\right\|_2^2.$$

Per-user aggregation and the worst 10% user mean reveal whether a good global mean hides systematic failures on a subset of series.

**Experimental goal.** Establish the no-adaptation reference, measure its temporal and user-wise heterogeneity, and provide the denominator against which every retrieval method must be compared.

## 2 Windows and chronological protocol

The integer $s$ is always the query date and the last observed date. The look-back and target are

$$X_s = (s-L, s] = \{z_{s-L+1}, \ldots, z_s\}, \qquad Y_s = (s, s+H] = \{z_{s+1}, \ldots, z_{s+H}\}.$$

They contain exactly $L$ and $H$ values; the look-back never contains the first forecast value. If a target interval begins at date $b$, its first query is $s = b-1$. The look-back may cross that boundary, but the complete $Y_s$ must lie inside the selected target interval. Consequently the held-out $T_3$ target dates do not shift when $L$ changes.

Raw dates are first split chronologically as $(T_0, T_1 + T_2, T_3) = (0.30, 0.50, 0.20)$. Extraction writes pooled `adapt` ($T_1$ $+T_2$) and held-out `eval` ($T_3$) payloads. Each downstream model makes its own whole-query-date split of `adapt`; users sharing one query date can never be divided between $T_1$ and $T_2$. The control evaluates $T_3$. Each query contains all users at one date, but the backbone is called as a univariate model with one channel per sample.

Dataset preparation automatically reads the sibling `config.json`. Portable top-level options and adaptation-scoped options are combined before the chronological split; user exclusions are additive and the selected config path is logged. The same preparation contract is used by extraction, the univariate control, and all downstream adapters. ETT, Weather, and Exchange Rate retain every non-date variate; constant-window audits do not remove their scarce channels.

The test profile uses Electricity 504–168 with sparse evaluation. Full widens the Chronos control to eight homogeneous-panel datasets and uses the same three look-back–horizon settings as the retrieval screen. The four additional ETT panels separate temperature from load at hourly and 15-minute cadences. Original ETTh1, ETTh2, ETTm1, ETTm2 and Weather are excluded because their channels mix unlike physical quantities; they are reserved for the retrieval study's mixed-quantity ablation. Ultra has the same direct Chronos control because this reference launcher has no second backbone.

The current pipeline produces this reference inside every retrieval extraction, so it shares the exact backbone, target windows, and evaluation dates used by the adapted candidates. There is no separate Slurm front.
**Implementation:** `src/experiments/extraction.py`
**Launchers:** `extraction.slurm`, `screen.slurm`, and `benchmark.slurm`

It writes window-level losses, a JSON summary, a tensor payload, error distributions, horizon-error curves, user scatter plots, and prediction examples. The primary comparison is the same $T_3$ MSE/nMSE later used by adapted methods.

| Component | Current sweep value |
|---|---|
| Datasets | Electricity, Traffic, Solar, Exchange, four separated ETT panels |
| Look-back–horizon | 168–24, 336–48, 504–168 |
| Backbone | Chronos |
| Input normalization | Instance normalization |
| Evaluation stride | 128 dates |
| Seed | 1 |

## 3 Interpretation

This experiment is descriptive rather than trainable. Any adapted method must beat the control on the same queries and aggregation rule. Improvements on $T_1$ or $T_2$ alone are insufficient evidence because those periods are used to fit adapters or gates.